

Edge Vision AI Co-Processing for Dynamic Context Awareness in Mixed Reality

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ABSTRACT

Spatial computing is evolving towards leveraging data streaming for computationally demanding applications, facilitating a shift to lightweight, untethered, and standalone devices. These devices are ideal candidates for co-processing, where real-time scene context understanding and low-latency data streaming are fundamental for general-purpose Mixed Reality (MR) experiences. This poster demonstrates and evaluates a scalable approach to augmented contextual understanding in MR by implementing edge AI co-processing through a Hailo-8 AI accelerator, a low-power ARM-based single board computer (SBC), and the Magic Leap 2 AR headset. The resulting inferences are streamed back to the headset for spatial reprojection into the user's vision.

1 INTRODUCTION

The fusion of artificial intelligence (AI) with mixed reality (MR) technologies is transforming how we perceive and interact with digital information in the physical world. As MR devices become lightweight and standalone, the demand for enriched, context-aware experiences grows. However, standalone MR headsets face significant computational constraints, hindering their ability to perform complex tasks such as occlusion-based semantic segmentation or dynamic environment redecoration. Real-time processing of these rich datasets is essential for next-generation MR applications, yet current hardware struggles to meet these demands [5].

Although edge computing offers a promising solution, MR systems still face challenges such as high latency, inefficient data handling, and difficulties in synchronizing virtual and real-world elements. Addressing these requires a system that minimizes latency, optimizes resource use, and adapts to dynamic environments. We therefore present a novel edge AI coprocessing system to enhance dynamic context awareness in MR, deploying an end-to-end AI processing pipeline that improves spatial cognition and supports dynamic interactions in complex settings. A glimpse of our platform is shown in Figure 1.

The contributions of this work can be summarized as follows:

- **Design and implementation of an edge AI co-processing platform:** We develop a framework for real-time vision processing of streamed camera data via WebRTC.
- **Demonstrative deployment of multiple vision AI models:** We deploy a suite of quantized vision AI models (YOLOv8s, CenterPose, FastDepth, and SCRFD + Arcface) to run simultaneously on the edge accelerator, demonstrating multi-modal analysis of the environment.

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- **Enhanced MR reprojection techniques:** We implement an improved co-planar reprojection method that aligns inference data with the real-world environment by combining inference data with camera extrinsics and AI depth estimation.

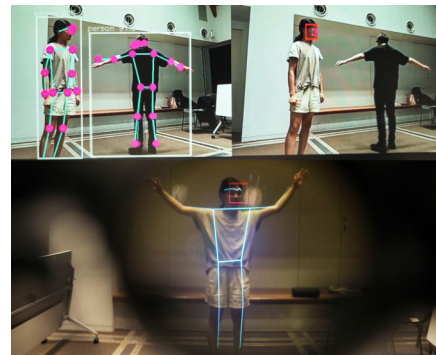


Figure 1: Visualization of pose estimation and facial recognition pipelines running on the SBC and on the Magic Leap 2

2 SYSTEM ARCHITECTURE

The system was designed to address, using easily accessible general-purpose hardware, intense per-frame computations at the lowest possible power consumption and with minimal latency. The architecture prepares video frames for AI inferencing and offloads these tasks to an AI accelerator. Below, we describe the hardware configuration and software pipeline of our system in detail.

2.1 Hardware Configuration

The platform centers around the Rockchip RK3588 SoC, an octa-core processor with integrated neural processing capabilities (6 TOPS for INT8). The RK3588 was chosen for its Raster Graphics Engine (RGA), which provides dedicated hardware acceleration for essential video processing tasks. Coupled with the Hailo-8 AI accelerator, which offers 26 TOPS of additional AI compute, the system supports concurrent execution of multiple AI models. Connectivity options include gigabit Ethernet and WiFi6, ensuring flexibility for various deployment scenarios.

2.2 Software Pipeline

The software stack combines open-source frameworks and custom solutions for seamless data flow. AIORTC, a Python-based WebRTC implementation, streams video and sensor data from the Magic Leap 2 headset to the edge processor. The GStreamer framework manages preprocessing and inference, leveraging Hailo-native plugins and the RGA for hardware-accelerated operations.

AI models—including YOLOv8s for object detection [4], CenterPose for pose estimation [7], FastDepth for depth estimation [6], and face detection and recognition (SCRFD [3] and ArcFace [1])—are

executed concurrently using the Hailo-8's round-robin scheduler. Custom C++ postprocessors convert raw inference outputs into structured data (e.g., bounding boxes, pose vectors), which are managed through Node-RED for real-time interaction and integration with external systems.

2.2.1 2D-to-3D Reprojection

AI-driven depth estimation maps 2D detection results into 3D coordinates. Depth data and camera extrinsics from the Magic Leap 2 are shared via the POSIX shared memory API, ensuring virtual objects align accurately with real-world elements. This 3D data, combined with object classification and detection confidence, is serialized into JSON format and passed through the system for further processing.

3 EVALUATION AND RESULTS

The testing methodology evaluated the system's efficiency and scalability, including a real-world deployment at Interop Tokyo 2024, where five AI models (YOLOv8s, CenterPose, FastDepth, SCRFD, and ArcFace) were demonstrated running simultaneously. However, challenges with the Hailo-8 scheduler caused instability with five models, making reliable benchmarking impractical. As a result, the presented results focus on tests with up to three models, assessing power consumption, photon-to-photon latency, and pipeline delay over Ethernet and WiFi connections.

3.1 Results

3.1.1 Power Consumption

The power consumption was measured for each model individually and for all three models running concurrently. The results demonstrate the system's energy efficiency under concurrent workloads, highlighting a power-scaling benefit when running simultaneous models.

Table 1: Power Consumption (W)

Configuration	Power (W)	Std Dev (W)
YOLOv8s	1.233	0.020
SCRFD_10g	1.134	0.0058
CenterPose_RepVGG_A0	1.102	0.0036
All Models RR@30FPS	2.631	0.040

3.1.2 Photon-to-Photon Latency

Photon-to-photon latency, the delay from image capture on the Magic Leap 2 (ML2) to displaying processed results, was tested over Ethernet and WiFi connections. The ML2 captured a timer displayed on a desktop PC and the transmitted video stream was displayed back on the PC. Timing differences were recorded at 120 frames per second.

Table 2: Photon-to-Photon Latency (ms)

Configuration	Latency (ms)
Depth + CenterPose (Ethernet)	167
All Models (Ethernet)	181
All Models (WiFi)	205

3.1.3 Pipeline Delay

Pipeline delay was measured for both standalone models and multi-model configurations. A base video pipeline delay of 9.449ms (std dev 7.25ms) was factored into all tests and should be added to each model's delay value to determine the overall pipeline delay. In the case of the multi-model configuration, each pipeline operates in parallel.

Table 3: Standalone Pipeline Delay (ms)

Model	Delay (ms)	Std Dev (ms)
YOLOv8s	13.764	6.560
CenterPose_RepVGG_A0	17.827	7.571
SCRFD_10g	9.024	3.984

Table 4: Multi-Model (simultaneous model) Pipeline Delay (ms)

Model	Delay (ms)	Std Dev (ms)
YOLOv8s	52.950	37.352
CenterPose_RepVGG_A0	41.362	24.978
SCRFD_10g	31.147	29.764

4 CONCLUSION

The proposed system offers notable advantages in versatility, reprojection accuracy, and efficiency. The multi-modal AI processing pipeline enables simultaneous execution of AI models, such as object detection and depth estimation, on the Hailo-8 accelerator, enhancing its adaptability for diverse MR applications. AI-driven depth estimation improves spatial coherence, while hardware accelerators like the Hailo-8 and RK3588's RGA ensure low power consumption and minimal latency, critical for real-time MR interactions.

However, some limitations must be addressed. High bandwidth usage between the Magic Leap 2 and Rock 5B SBC could hinder performance in network-constrained scenarios, and scalability for multi-device environments remains untested, leaving open questions about its performance in more complex, collaborative settings involving multiple MR headsets or devices.

Potential future enhancements include adopting split learning approaches, such as *DynSplit* [2], to reduce bandwidth by running parts of AI inference locally and transmitting compressed intermediate data to the edge or cloud. Larger-scale, collaborative deployments and scenario-specific evaluations are necessary to further understand the system's scalability, robustness, and practical applicability.

4.1 Acknowledgments

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